

Adrian Martinez

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SUMMARY

B.S. Electrical Engineering candidate. Conducted a year-long analytical research project in materials science at Argonne National Laboratory. Won 3 leadership awards over 2 consecutive years. Placed 1st in a 30-team electrical and computer engineering programming competition.

EDUCATION

Illinois Institute of Technology, Chicago, IL
Bachelor of Science, Electrical Engineering

MAY 2029

- Dean's List (All Semesters)

SKILLS

- **Hardware:** Circuit assembly | Sensor integration | 3D Printing |
- **Software:** C++ | Java | HTML & CSS | SolidWorks |
- **Laboratory:** Iterative Prototyping | Design Validation Testing | Troubleshooting & Debugging |
- **Languages:** Illinois State Seal of Biliteracy in English and Spanish |

WORK EXPERIENCE

MECHANICAL DESIGN ENGINEERING INTERN

JAN 2026 – APR 2026

Cupp Golf, Remote

- Contributed to the research and development for a golf training aid designed for maintaining a consistent swing radius.
- Utilized 3D CAD modeling to design and iterate a quick-release clamp component, modifying the fastening configuration to improve adjustability, strength, and repeatability while considering manufacturing practicality and user ergonomics.
- Performed design validation through comparative prototyping, analyzing how structural geometry influences device performance and consistency.

EXEMPLARY STUDENT RESEARCH PROGRAM (ESRP) - TEAM LEADER

SEPT 2024 – APR 2025

Argonne National Laboratory, Lemont, IL

- Selected out of 1,500 students to represent Neuqua Valley High School in the ESRP Cross-Team Committee, a leadership group for participating schools to discuss their research.
- Led a multidisciplinary research team of 8 to study the strength and flexibility of nylon carbon fiber and acrylonitrile styrene acrylate (ASA) using tensile testing to identify potential alternative materials for prosthetic components.
- Analyzed data from X-ray diffraction and X-ray absorption fine structure spectroscopy and concluded that nylon carbon fiber reveals a more organized and crystalline atomic structure, exhibiting promising qualities suitable for prosthetic ankle-foot systems.

PROJECT EXPERIENCE

Photography Website Design:

JUN 2026 – PRESENT

- Built a website using HTML and CSS to showcase my photography.
- Designed a responsive layout with a navigation menu, image galleries, and a contact form.

Accessible Pedestrian Signal (APS) & Traffic Light System:

OCT 2025 – DEC 2025

- Programmed a pedestrian call-button that triggers visual and auditory cues which simulate a real-world traffic intersection.
- Wired hardware components on a breadboard, including a buzzer, button, LCD screen, resistors, and LEDs.

Autonomous Line-Following Robot:

AUG 2025 – SEP 2025

- Designed and programmed an autonomous robot capable of detecting and navigating a marked path using infrared sensors.
- Executed multiple iterative testing and troubleshooting phases to increase tracking accuracy.
- Won 1st place in a 30-team competition by setting a record for the fastest course completion time.